2022

STATISTICS — HONOURS

Paper : CC-5

(Linear Algebra)

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer ***any five*** questions : 2×5

(a) Verify that $W = \{(a, b, c, d) : \text{with } c = a \text{ and } d = a + b\}$ is a vector subspace.

(b) A set of vectors consisting a null vector can not be independent— Justify.

(c) Let $<A>$ be the subspace spanned by the set of vectors A.

If $A_1 = \{(1,1,0,0)'; (1,0,1,1)'\}$ and $A_2 = \{(2, -1, 3, 3)'; (0,1,-1,-1)\}$, then verify whether or not $<A_1> \equiv <A_2>$.

(d) Find A^{-1} if $A^2 - A + I = 0$.

(e) Let A be an $n \times m$ matrix of rank r and B be another matrix of order $r \times m$ with rank $(m-r)$ such that $AB' = 0$. Show that $A'A + B'B$ is of rank m.

(f) Show that the characteristic roots of an idempotent matrix are either zero or one.

2. Answer ***any two*** questions : 5×2

(a) If A is an $n \times n$ matrix $((a_{ij}))$, with $a_{ij} = P^{|i-j|}$ $(0 < P < 1)$, then show that $|A| = (1-P^2)^{n-1}$. Also find the inverse of A.

(b) Define column space of a matrix. Consider the real matrix.

$$A = \begin{bmatrix} 4 & 1 & 6 & 0 \\ 0 & 1 & 2 & -4 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Now denoting by $e(A)$, the column space of A, show that

$$e(A) = \{(a,b,c)' : a - b - 4c = 0\}.$$

(c) Show that characteristic values of real symmetric matrix are real while those of a skew symmetric matrix are imaginary.

3. Answer ***any three*** questions : 10×3

(a) (i) Define inverse of a matrix. Find inverse of the matrix.

$$\begin{bmatrix} a & b & b\cdots b \\ b & c & d\cdots d \\ b & d & c\cdots d \\ \vdots & \vdots & \vdots \quad \vdots \\ b & d & d\cdots c \end{bmatrix}^{n\times n}$$

(ii) For a real square matrix A, if $(I+A)$ is non-singular and $(I+A)^{-1}(I-A)$ is skew symmetric, show that A is orthogonal.

(b) (i) Find system of two independent linear homogeneous equations in x_1, x_2, x_3, x_4 such that

$(x_1, x_2, x_3, x_4) = (1, 2, 3, 4)$ and

$(x_1, x_2, x_3, x_4) = (2, 3, 4, 1)$ are solution of the system.

(ii) If A and B are square matrices of same order, then show that

Rank $(AB - I) \leq$ Rank $(A - I)$ + Rank $(B - I)$.

(c) (i) Define null space and nullity of a matrix. State and prove necessary and sufficient condition for a system of non-homogeneous equations $B^{m\times n}x^{n\times 1} = P^{m\times 1}$ to have a solution.

(ii) If $A^{m\times n}$ and $B^{n\times p}$ are two matrices, then prove that rank $(AB) \geq$ rank(A) + rank $(B) - n$

(d) (i) Find the eigenvalues of the matrix

$$A = (\alpha - \beta) I_p + \beta\, I_p I_p'$$

(ii) For a non-null vector u, let $A = uu'$. Show that the eigenvalues of A are 0 and $u'u$.

(e) (i) Derive a set of sufficient conditions for a square matrix to be non-negative. Indicate one statistical application of this result.

(ii) Let A be a real symmetric matrix of order n with eigenvalues $\lambda_1, \lambda_2 ... \lambda_n$, then show that

$$\max_{x\neq 0} \frac{x'Ax}{x'x} = \max(\lambda_1, \lambda_2 ... \lambda_n).$$